



CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

Lecture No.- 03



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Recap of Previous Lecture



Topic

Logic Part-02



Topics to be Covered



Topic

Logic Part-03



(Simplification)



Type 2 → logically equivalence:

$$1) \quad p \wedge p \equiv p$$

$$p \vee p \equiv p$$

$$2) \quad p \vee T \equiv T$$

$$p \wedge F \equiv F$$

$$3) \quad p \wedge T \equiv p$$

$$p=T \quad T \wedge T \equiv T$$

$$p=F \quad F \wedge T \equiv F$$

$$4) \quad p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

$$5) \quad \begin{aligned} p \wedge (q \wedge r) &\equiv (p \wedge q) \wedge r \\ p \vee (q \vee r) &\equiv (p \vee q) \vee r \end{aligned} \quad \left(\begin{array}{l} \text{Associative} \\ \downarrow \\ \text{operators are same} \end{array} \right)$$

$$p \vee F \equiv p$$

$$\begin{aligned} p=T & \quad T \vee F \equiv T \\ p=F & \quad F \vee F \equiv F \end{aligned}$$

Distributive law: (operators are diff)

$$a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$$

$$a \wedge (b \vee c) \equiv (a \wedge b) \vee (a \wedge c)$$

$$a \vee (b \vee c)$$

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$

$$a \cup (b \cap c) = (a \cup b) \cap (a \cup c)$$

$$a \cap (b \cup c) = (a \cap b) \cup (a \cap c)$$

logic.	Digital logic.	set
T	1	$\mathcal{U}$
F	0	$\emptyset$
$\wedge$	$\cdot$	$\cap$
$\vee$	$+$	$\cup$



Topic : Logic



** absorption law :

$$a \vee (a \wedge b) \equiv a.$$

$$a = T$$

$$a \vee (a \wedge b)$$

$$T \vee (\text{---})$$

$$T$$

$$a = F$$

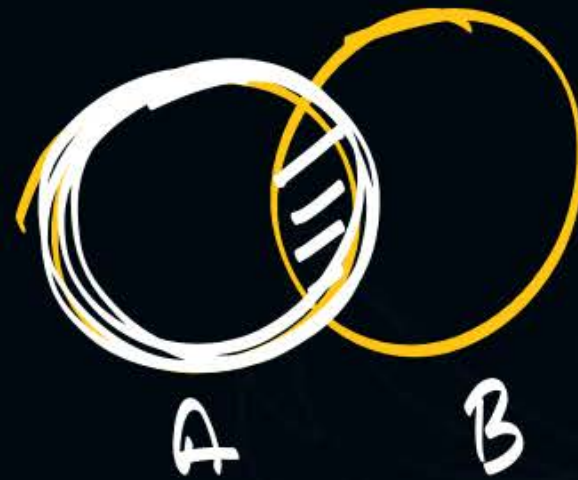
$$a \vee (a \wedge b)$$

$$F \vee (F \wedge \text{---})$$

$$F \vee F$$
$$F$$

Set :

$$A \cup (\underline{A \cap B}) \equiv A.$$



$$a \vee (a \wedge b) \equiv a, \quad a \wedge (a \vee b) \equiv a.$$

what is outside is
inside
& operators are diff.

$$a \vee (a \wedge (b \rightarrow c)) \equiv a.$$

$$a \vee (a \wedge ((b \wedge c) \rightarrow d)) \equiv a.$$

$$a \wedge (a \vee b) \equiv a.$$

$$(a \rightarrow b) \vee ((a \rightarrow b) \wedge (c \rightarrow d)) \equiv (a \rightarrow b)$$

~~Good's
Rule.~~

$$a \rightarrow b \equiv \neg b \rightarrow \neg a.$$

$$a \rightarrow b \equiv \neg a \vee b.$$

$$\begin{aligned}\neg(a \rightarrow b) &= \neg(\neg a \vee b) \\ &= a \wedge \neg b.\end{aligned}$$

$$\neg(a \rightarrow b) \equiv a \wedge \neg b.$$



De Morgan's Law

$$\neg(a \vee b) \equiv \neg a \wedge \neg b.$$

$$\neg(a \wedge b) \equiv \neg a \vee \neg b.$$

$$(a \rightarrow b) \wedge (a \rightarrow c) \equiv a \rightarrow (b \wedge c)$$

$$(\neg a \vee b) \wedge (\neg a \vee c)$$

$$\neg a \vee (b \wedge c)$$

$$a \rightarrow (b \wedge c)$$

$$(a \rightarrow c) \wedge (b \rightarrow c) \equiv (a \vee b) \rightarrow c$$

$$(\neg a \vee c) \wedge (\neg b \vee c)$$

$$(\neg a \wedge \neg b) \vee c$$

$$\neg (a \vee b) \vee c$$

$$(a \vee b) \rightarrow c$$

$$a \rightarrow b \equiv \neg b \rightarrow \neg a.$$

$$a \rightarrow b \equiv \neg a \vee b.$$

$$\left\{ \begin{array}{l} (a \rightarrow b) \wedge (a \rightarrow c) \equiv a \rightarrow (b \wedge c) \\ (a \rightarrow c) \wedge (b \rightarrow c) \equiv (a \vee b) \rightarrow c. \end{array} \right.$$

$$\left\{ \begin{array}{l} (a \rightarrow b) \vee (a \rightarrow c) \equiv a \rightarrow (b \vee c) \\ (a \rightarrow c) \vee (b \rightarrow c) \equiv (a \wedge b) \rightarrow c. \end{array} \right.$$

$$a \leftrightarrow b \equiv (a \rightarrow b) \wedge (b \rightarrow a)$$

$$\equiv (\neg a \vee b) \wedge (b \rightarrow a)$$

$$\equiv (a \rightarrow b) \wedge (\neg b \vee a)$$

$$\equiv (\neg a \vee b) \wedge (\neg b \vee a)$$

$$p \wedge (\neg r \vee \boxed{q \vee \neg q}) \vee (\textcircled{r} \vee t \vee \textcircled{\neg r}) \wedge \neg q$$

$$p \wedge (\neg r \vee \underline{\underline{I}})$$

$$\frac{p \wedge I}{p}$$

↓

$$(\underline{\underline{I}} \vee t) \wedge \neg q$$

$$\frac{I \wedge \neg q}{\neg q}$$

- a) $p \wedge \neg q$
- ☒ b) $p \vee \neg q$
- c) t
- d) $p \rightarrow \neg q$

$$\boxed{p \vee (p \wedge q)} \vee (p \wedge q \wedge \neg r) \wedge (p \wedge \neg r \wedge t) \vee \underline{t}$$

$$\frac{\underline{p} \vee (p \wedge q \wedge \neg r)}{p}$$

$$\wedge$$

$$\frac{t \vee (p \wedge \neg r \wedge t)}{\underline{t} \vee (\underline{t} \wedge p \wedge \neg r)} \\ t$$

- ☒ a) $p \wedge t$
- b) $q \wedge t$
- c) $p \wedge \neg r$
- d) $p \wedge q$

GATE-08

I. $p \vee \neg q$

II. $\neg(\neg p \wedge q)$

III. $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$

IV. $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

$$(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$$

$$p \wedge (q \vee \neg q)$$

$$\frac{p \wedge T}{p \vee (\neg p \wedge \neg q)}$$

equivalent

A) I & II.

B) I, II & III

C) I, II & IV.

D) all.

GATE-08

I. $p \vee \neg q$.

II. $\neg(\neg p \wedge q)$.

III. $(\boxed{p \wedge q}) \vee (\boxed{\neg p})$.

IV. $(p \wedge q) \vee (p \wedge$

$(\boxed{p \wedge q}) \vee (\boxed{p \wedge \neg q}) \vee (\neg p \wedge \neg q)$

$p \wedge (\underline{q \vee \neg q})$

$\frac{p \wedge \tau}{p \vee (\neg p \wedge \neg q)}$

$(\underline{p \vee \neg p}) \wedge (p \vee \neg q)$

$\underline{\tau \wedge (p \vee \neg q)}$

$p \vee \neg q$.

equivalent

A) I & II.

☒ B) I, II & III

C) I, II & IV.

D) all.



2 mins Summary

Topic

One

Logic (Part -03)

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU